RIANA MOODLEY

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PROFESSIONAL PROFILE

Final-year Electrical Engineering student at the University of Zululand, graduating end of 2026, and South32 bursary holder. Hands-on experience in PLC automation, power conversion and system simulation, including a swing-powered energy harvesting system, a CODESYS batch-mixing control system and a smart traffic lighting system. Seeking a graduate-in-training position in automation, power systems or telecommunications, with plans to register with ECSA as a Candidate Engineer.

PROFESSIONAL ACTIVITIES & ACHIEVEMENTS

South32 Bursary
Bursary Recipient
- Awarded a bursary on academic merit.

South African Institute of Electrical Engineers (SAIEE) *2024 – Present*
Student Member
- Active student member, engaging with the engineering professional community and staying current with industry developments.

Cars4Mars *2025*
Participant
- Took part in the Cars4Mars 2025 programme.

EDUCATION

University of Zululand *2023 – Nov 2026*
Bachelor of Engineering, Electrical Engineering (Richards Bay Campus)
- Selected study areas: process control and instrumentation, telecommunications, power electronics and machines, power systems, and final year project.

Hoerskool Richardsbaai *2022*
Secondary School (Matric), Average: 70.28%
- Subjects: Mathematics, Physical Sciences, Life Sciences, Engineering Graphics and Design.

PRACTICAL INVOLVEMENTS & CERTIFICATIONS

1. Eskom Expo for Young Scientists
Judge
- Judged young scientists' projects, assessing their technical work and presentation.
- Gave participants advice on how to improve their projects.

2. Short Courses: SMR Machinery and Radiation Protection
Completed short courses in small modular reactor (SMR) machinery and radiation protection, and received radiation protection certificates.

3. Huawei HCIA-Artificial Intelligence (HCIA-AI) Certification, Huawei ICT Academy *2026 – Present*
Selected for the certification programme covering artificial intelligence fundamentals, machine learning concepts and Huawei's AI development ecosystem.

TECHNICAL SKILLS

Industrial Automation: CODESYS, WPLSoft, PLC programming, sensor-based control logic, PID control, instrumentation.

Control & Simulation: MATLAB/Simulink, PSpice, PowerWorld Simulator, PSCAD, system modelling.

Power & Electronics: Boost/buck converters, thyristor rectifiers, DC motor drives, load flow studies, protection schemes.

Telecommunications: Cellular network engineering, Okumura-Hata propagation modelling, Erlang-B traffic analysis, satellite link budgets, fibre-optic systems.

Engineering Documentation: Technical reports, data analysis and presentations using Microsoft Word, Excel and PowerPoint.

ENGINEERING & TECHNOLOGY

Swing-Powered Electrical Generator - Final-Year Project *2026*

- Led the design of a kinetic energy harvesting system that converts swing motion into usable electrical power, from concept through to system modelling.
- Integrated a DC motor generator, freewheel/roller-clutch mechanism and boost converter, with battery charge management.
- Modelled the power conditioning system in Simulink, aiming for steady output across variable swing speeds, with validation through simulation and physical prototyping.

Fertilizer Batch Mixing Automation *2026*

- Sole developer of the complete PLC control system, from logic design through to implementation, using CODESYS and WPLSoft.
- Programmed ingredient ratios, mixing sequences and multi-stage timing to give repeatable, consistent batches.

Smart Traffic Lighting System - Group Project *2025*

- Owned the lighting configuration in a team design of an intelligent traffic management system, coordinating with team members to deliver an integrated solution.
- Programmed sensor-based input logic and automated control sequencing in CODESYS, reducing intersection congestion and improving traffic flow.

ADDITIONAL INFORMATION

Driver's licence: Code 8.

Interests: Taekwondo (Green Belt), strength training and running.

REFERENCES

Dr. Franck Cinyama Mushid from University of Zululand

- Mushidc@unizulu.ac.za